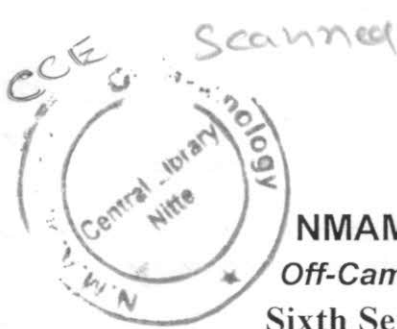




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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

CC2104-1- PRINCIPLES AND PRACTICES OF SOFTWARE ENGINEERING
(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **In the Waterfall Model, when is testing performed?**
A) After each phase
B) At the end of development
C) Before the design phase
D) Continuously throughout the process
2. **The Incremental Model is best suited for:**
A) Large projects with well-defined requirements
B) Small projects with evolving requirements
C) Hardware-intensive projects
D) Non-software projects
3. **Which of the following is NOT a core software engineering process activity?**
A) Software specification
B) Software testing
C) Software maintenance
D) Software elimination
4. **What is the main reason for software project failure?**
A) Poor communication among stakeholders
B) Use of outdated programming languages
C) Too much software documentation
D) Fast software delivery
5. **What is the primary purpose of the requirements engineering process?**
A) To develop software applications
B) To write program code efficiently
C) To define, document, and maintain requirements
D) To remove all project risks
6. **Prototyping in requirements elicitation is used to:**
A) Provide a visual representation of requirements
B) Speed up development without analysis
C) Finalize the design before implementation
D) Write test cases
7. **Which of the following is a key component of a Software Requirements Specification (SRS)?**
A) Source code implementation
B) User interface design
C) Functional and non-functional requirements
D) Programming language details
8. **What is the primary goal of requirements specification?**
A) To describe how the system will be implemented
B) To test the software application
C) To identify and fix bugs
D) To formally define what the system should do
9. **Which of the following best describes a state diagram?**
A) A model that shows how data moves through a system
B) A diagram that represents the static structure of a system
C) A model that captures the lifecycle of an object through various states
D) A diagram that represents user interactions with a system

10. Which of the following is NOT a key architectural design decision?
 A) Selection of programming language B) Organizational structure of the development team
 C) High-level system structure D) Technology selection
11. Which architectural view focuses on how components communicate with each other?
 A) Physical view B) Development view
 C) Logical view D) Process view
12. Which application architecture is best suited for distributed systems with independent components?
 A) Monolithic architecture B) Client-server architecture
 C) Microservices architecture D) Layered architecture
13. Which UML diagram is used to model the flow of interactions in a system?
 A) Activity diagram B) Sequence diagram
 C) Class diagram D) Component diagram
14. In a UML class diagram, which relationship represents inheritance?
 A) Solid line with a diamond B) Dashed line with an arrow
 C) Solid line with a hollow triangle D) Solid line without an arrow
15. Which of the following is a key characteristic of Agile development?
 A) Iterative and incremental development B) Heavy upfront documentation
 C) Strictly defined requirements before coding begins D) Rigid adherence to a single development methodology
16. What is the main goal of test-driven development (TDD) in XP?
 A) a) To create extensive project documentation B) To write tests before writing the actual code
 C) To delay testing until the final phase of development D) To minimize customer involvement in testing
17. What is the first step in risk management?
 A) Risk monitoring B) Risk control
 C) Risk mitigation D) Risk identification
18. Which of the following is a common team structure in software projects?
 A) Functional teams B) Matrix teams
 C) Agile teams D) All of the above
19. What is the primary purpose of project scheduling?
 A) To assign tasks randomly to team members B) To estimate time and resources needed for project completion
 C) To eliminate the need for risk management D) To determine software pricing
20. Which metric measures the number of independent paths in a program?
 A) Cyclomatic complexity B) Function point analysis
 C) Code coverage D) Test case execution rate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the differences between the Incremental Model and the Waterfall Model.	8	L*2	1	1,2
	b) Identify which of the three models (Waterfall, Incremental, or Spiral) would be best for a project with frequent revisions and evolving requirements and justify your choice.	8	L2	1	1,2
2.	a) What are the core components of a Patient Information System used in mental health care, and how do they support patient management and treatment?	8	L3	1	1,2
	b) With the neat diagram Explain the requirements engineering process.	8	L2	1	1,2

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|-------|--|---|----|---|-----|
| 3. a) | What is the difference between functional and non-functional requirements? Provide examples of each. | 8 | L2 | 2 | 1,2 |
| b) | Explain the Different Ways of writing a system requirements specification. | 8 | L2 | 2 | 1,2 |

Unit – II

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|-------|---|---|----|---|-------|
| 4. a) | What are structural models, and how do they differ from behavioral models in terms of system representation? | 8 | L2 | 3 | 1,2,3 |
| b) | Define interaction models and describe their role in software design. | 8 | L2 | 3 | 1,2,3 |
| 5. a) | What are architectural patterns in software engineering, and list at least four common types of architectural patterns used in software development? | 8 | L2 | 3 | 1,2,3 |
| b) | With a diagram explain the architecture of a packing robot control system. | 8 | L3 | 3 | 1,2,3 |
| 6. a) | Use a sequence diagram to represent the process of placing an online order in an e-commerce system. | 8 | L3 | 4 | 1,2,3 |
| b) | Given a scenario where a team is working under extreme time pressure, apply the principles of Extreme Programming (XP) to ensure quick and efficient development. | 8 | L2 | 4 | 1,2,3 |

Unit – III

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|-------|--|---|----|---|-------|
| 7. a) | Explain the difference between technical and non-technical risks in a software development project. | 8 | L2 | 5 | 1,2,3 |
| b) | Given a software product, apply cost estimation techniques to determine an appropriate pricing model for that product. | 8 | L2 | 5 | 1,2,3 |
| 8. a) | Explain the benefits of following a structured, plan-driven approach in large-scale software projects. | 8 | L2 | 5 | 1,2,3 |
| b) | Explain how a Gantt chart can help in visualizing the schedule and progress of a software project. | 8 | L2 | 5 | 1,2,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
